



SAFETY Department
SIMHADRI

Note No :1



Ref:SMTTPP/SAFETY/2023-24/NOTE/633974

Date:07/11/2023(15:35:34)

Subject:Approval of LMI on Hazard Identification and Risk Assessment -reg.

Subject:Approval of LMI on "Hazard Identification and Risk Assessment" -reg.

The LMI on Hazard Identification and Risk Assessment, Ref no. LMI - OPS-SAFT-001, Rev:02, Dated: 07.11.2023 is reviewed and changes incorporated in the LMI as follows:

In Annexure 1: Table Containing “EXAMPLE CASES: RATING – CONSEQUENCE / SEVERITY /HARM” is modified

This supercedes LMI/OD/OPS/SAFT/001, Dated: 21.09.2021.

Put up for approval of Competent authority.

Submitted by:

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Note No:2

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Note No:6

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Note No:16

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May pl forward.

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Note No:20

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Note No:21

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Note No:24

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Note No:27

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Note No:30

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Note No:31

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Note No:32

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Note No:33

May please be approved

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Note No:34

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On:13/12/2023(11:28:44) **Ref:**SMTPP/SAFETY/2023-24/NOTE/633974



SIMHADRI

NTPC LIMITED
SIMHADRI SUPER THERMAL POWER STATION

**Local Management Instruction –
LMI/OD/OPS/SAFT/001**

(Ref: COS-ISO-00-OD/OPS/SAFT/001- Rev. No. 02, 02 Oct 2023)

Revision: 02

Dt: 07.11.2023

HAZARD IDENTIFICATION & RISK ASSESSMENT

Approved for Implementation by.....

HEAD OF PROJECT (Simhadri)

Date: __.11.2023

SAFETY DEPARTMENT

TITLE OF LMI:		Hazard Identification and Risk Assessment of Functions at Simhadri Super Thermal Power Station.					
LMI No:		LMI/OD/OPS/SAFT/001					
ISSUE NO	REVISION NO	DATE	CHANGES DONE	PREPARED BY	CHECKED BY	REVIEWED BY	APPROVED BY
01	00	.07.21	All clauses are new as the document is being issued for the first time.	P S S REDDY	ALL HODs	GM (O&M)	HOP
02	02	.11.23	In Annexure 1: Table Containing “Example Cases: Rating – Consequence/Severity /Harm” is modified.	P S S REDDY	ALL HODs	GM (O&M)	HOP

NTPC LIMITED**SIMHADRI SUPER THERMAL POWER STATION****LMI APPROVAL DOCUMENT**

Title of LMI:	Hazard Identification and Risk Assessment of Functions at Simhadri Super Thermal Power Station	
LMI No.:	LMI/OD/OPS/SAFT/001	
Issue No: 0	Revision: 02	Date: 07 .11.2023

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Approved for Issue by	Sanjay Kumar Sinha	GROUP GENERAL MANAGER (S)	
Date of Issue			

Hazard Identification & Risk Assessment

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Hazard Identification & Risk Assessment

1.0 Summary

1.1 Superseded Documents: LMI-OD/OPS/SAFT/001 Rev. No.: 0

1.2 Scope:

This procedure applies to all activities and services carried out in all operational projects of NTPC.

All potential major emergency scenarios shall be assessed. While ideally all works need to be assessed for risks, the following works shall always be preceded by HIRA study: Work within confined spaces, Work at heights/elevated positions, Hot work, Trenching and excavation, Work involving/near hazardous substances, Work on or, near water, Work on or testing live electrical circuits, Work involving power tools, Driving involving heavy vehicles/load, Work on HV apparatus, Work involving high pressure & temperature, etc.

1.3 Review

Review Responsibility : Safety dept.

Review Frequency : 2 yearly basis, or as and when required, whichever is lower

Review Approval : Head of Safety will be reviewing authority and BUH will be approving authority for any revision in this document.

2.0 Introduction:

NTPC's Safety Policy states, among other things, that "our activities carry various hazards; however, all such hazards can be identified". Thus, effective identification of all hazards associated with the company's activities forms a part of company's Safety Policy and deemed essential to prevent accidents. This procedure describes the formal, structured process for identifying the occupational health and safety hazards associated with any process and assessing the risk level of consequences of those hazards. The results of risk assessment (HIRA) will be used as an input to establish and prioritize health and safety objectives, targets, and programs to achieve overall risk-reduction and facilitate continual improvement in occupational health and safety performance. Any record that is maintained as an electronic copy shall be in a non-editable format like a secure PDF.

This OD does not contain elements that can vary as per location. Hence no modification or, adaptation for location is expected for this OD.

3.0 Revision Details

In Annexure 1: Table Containing "EXAMPLE CASES: RATING – CONSEQUENCE / SEVERITY /HARM" is modified.

5.0 Roles & Accountability

1. Head of the Project/Stations (HOP) is accountable for:
 - a. Ensuring identification of all hazards and assessment of risks for all the project's processes (HIRA) is in place;
 - b. Appoint of Risk Assessors for hazards identification and risk assessment.
 - c. Ensuring review of the HIRA register on a yearly basis.
2. Heads of Sections (HOS) are accountable for:
 - a. Persons in their Section are aware of their functions HIRA.
 - b. Ensuring that the respective Engineer-in-charge (EIC) shall spot, report and deal with hazards to help create safe and incident free workplace.
 - c. Maintaining records of HIRA, for their respective functions.
 - d. Reviewing and updating HIRA register on a yearly basis.

For projects where operations and maintenance functions are outsourced to third-parties, the designated Person accountable for the performance of such a project, where ever he/she is located, shall be accountable for the above two roles.

3. EICs are accountable for carrying-out hazard identification and risk assessment for all hazards associated with their respective works.
4. Site Safety Department is accountable for
 - a. Ensuring HIRA review is a part of periodic Safety Committee Meetings.
 - b. Identification of all resources to facilitate the HIRA process.

6.0 Requirements**6.1 Appointment of Risk Assessors**

Each HOS will nominate persons from within their Section to undertake the duties of a risk assessor. These persons will receive training in the area of assessment techniques and upon demonstrating the following abilities, will be appointed by the HOP as Risk Assessors.

Areas of competence for appointment as Risk Assessor:

Possess knowledge of the work, plant, apparatus, methods, equipment, procedures, personnel and workplace.

Have a good balance of practical skills, theoretical knowledge and actual experience of relevant systems.

Ability to make an assessment of the problem and prioritize any actions.

Can communicate/discuss the assessment with the person doing the task/work.

Assessment of this competence will be made by the HOS, Head (O&M/Project) and where required, by the HOP and/or external specialist support.

Refresher training of appointed Risk Assessors will be provided once every two years, preferably using standard training modules for consistency of concept. Training & assessment records of all appointed assessors shall be maintained.

6.2 The Assessment Process

A risk assessment will be 'adequate' or 'suitable and sufficient' and will be based on the judgment of whether the hazards are significant and whether they are covered by precautions which mean that the risk is small.

The HOS will ensure that **all activities** under their control are listed out in a HIRA register. Such a register can be a hard copy or, maintained as a soft (electronic) copy. When maintained as a soft copy, control must be in place such that unauthorized changes to the document are not possible. All Sectional activities shall be assessed and where it is felt the risk is significant, a formal assessment is undertaken by an appointed assessor and the findings formally recorded using the form as given in Annexure-2. Where it is agreed that the risks are insignificant and a formal assessment is not required, a note will be made in the HIRA register confirming this and recorded in the HIRA register controlled by that HOS. Guidance on the Hazard Id & Risk Assessment approach is given in Annexure-1.

When a formal risk assessment has been completed it will be reviewed by the assessor, HOS and if necessary, the Head (O&M) and where residual risk warrants, by the HOP; to agree on the additional precautions that need to be taken to minimize risk to persons. Where this need requires a formal method of work to be agreed, then the same will be prepared and implemented to manage the risk to persons. If the assessment identifies a need for PPE, special tools or equipment and training for persons then these requirements will be actioned through the responsible HOS, before the activity continues. All such training records shall be maintained.

6.3 Assessment Records

All assessments, whether a formal assessment or, a note stating why an assessment was not required will be recorded and maintained in a HIRA register controlled by the respective HOS. Each assessment will have a unique identification number. The details of which will be recorded in the HIRA register. Formal assessments will be recorded on the form as given under Annexure-2.

HIRA Assessments will be reviewed when it is 'suspected that the assessment is no longer valid' or, there has been a 'significant change'. Review of assessments will also be made where the management of risk is dependent on continued effectiveness of control measures. In any case, there shall be an annual review of the HIRA register by the HOS. Records of such reviews shall be maintained by the respective HOS.

6.4 Work Execution and Responsibilities

Where the precautions identified in the risk assessment are to be implemented by persons involved in the work, such persons will be suitably informed of these requirements by the EIC and records maintained that they understand such precautions. If necessary, these requirements will be recorded in writing and issued to the person in the form of a method statement or, a work instruction. EIC shall be responsible to ensure any person set-off to work is trained to do so and is aware of all hazards related to that work.

7.0 Records

Any record that is maintained as an electronic copy shall be in a format that cannot be edited. The following records shall be maintained:

HOP:

- HIRA for the project – **Responsibility: Safety Department**
- List of Risk Assessors of the project with appointment & acceptance Records – **Responsibility: Head of Sections.**
- Training records on HIRA – **Responsibility: HR-RLI**

All HOSs shall maintain:

- HIRA register for their functions;
- training records of their Section employees on HIRA for their area;
- evidence of a system for checking implementation of the control measures;
- evidence of corrective action (including exemplary penal action) taken when control measures were not implemented.

Responsibility: All Heads of Section

8.0 Audit

Evaluation of the compliance with this LMI shall be carried out at least once annually. The compliance evaluation assessment/audit can be included as part of other compliance audits or assessments like technical compliance audits.

The audit shall be a combination of verification of records, verification of implementation in the field and interviews.

System: Auditors shall validate the evidence of a system to verify implementation of risk control measures.

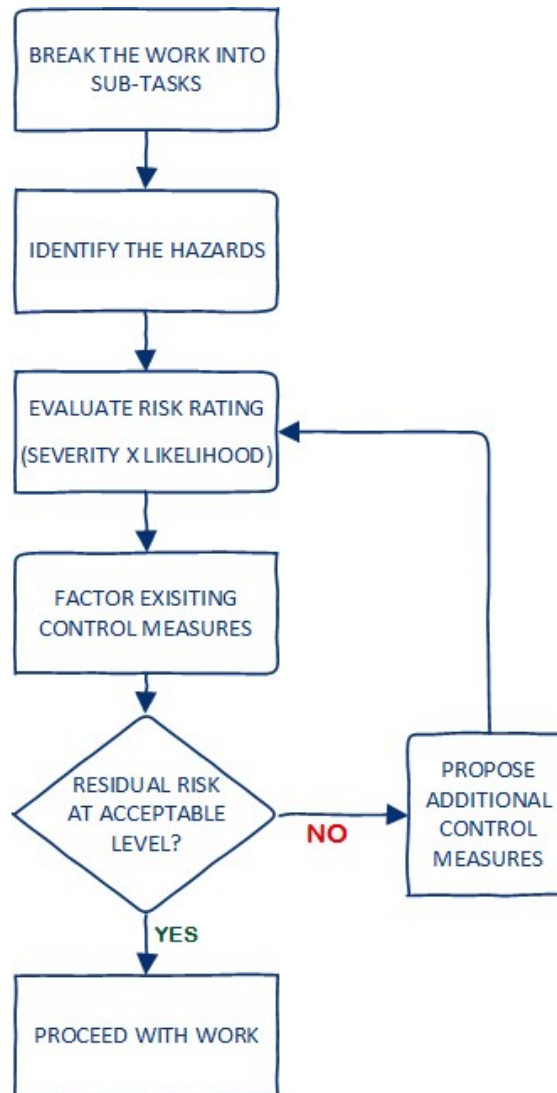
Records: Audit shall include verification of all the records detailed under 7.0.

Implementation: Field audit shall be in the form of random assessments. Auditor shall take a random sample of at least five activities from at least three different functions, including one non-technical function to assess the implementation of the committed control measures. Evidence of action initiated, including exemplary penal action, for failure to implement risk control measures shall also be verified and included in the audit report.

Interviews: check with at least five employees randomly selected from different functions on their awareness of HIRA of their respective function; HOS of respective function shall be accountable for demonstrating compliance to implementation and awareness requirements. Activities verified and persons interviewed shall be part of the audit report.

For projects where operations and maintenance functions are outsourced to third-parties, the designated Person accountable for the performance of such a project, where ever he/she is located, shall be accountable for ensuring periodic audits of the system at such projects under his/her control and maintain records.

Annexure 1 - Process For Hazard Identification & Risk Assessment

**Consider what should be assessed and prioritized**

The main areas can be considered for assessment is:

- Routine, non-routine, controlled or uncontrollable activities, as well as impacts, undesired effects associated with emergency scenarios.
- Activities of all persons having access to the workplace, incl. contractors & visitors.
- Human behavior, capabilities and other human factors.
- Change(s), including planned or new developments, and modifications.
- Work procedures including their consideration of human capabilities
- Reasonably foreseeable potential emergency situations.

Examples of assessable areas:

Activities	e.g. work at height, major overhaul.
Materials / Substances	e.g. chemicals, flammable aerosols.
Equipment machine.	e.g. portable grinder, welding
Workplace	e.g. boiler, fuel oil area.
People	e.g. helper.
Procedures	e.g. * methods of work. * supervision. * contingency plans for failures.

The priority can be decided by reference to factors such as:

- Is this a new job / process?
- Does statistical accident data indicate problems?
- Has an accident investigation revealed problems?
- Has a Safety Representative or Safety Committee highlighted a problem?
- Have we had a near miss (hit)?

Break the task into sub-tasks

All jobs that would necessitate a Risk Assessment are made up of smaller sub-tasks. For example, work at height would require arranging access platforms/scaffolds that need to be inspected before use etc. Break the task in to discrete sub-tasks that can be assessed.

Identify Hazards

The success of a risk assessment depends on a practical and comprehensive identification of hazards. Assessors should:

- ask questions such as "What could cause deviations from safe working procedures or conditions and what are likely to be the resulting harms or consequences"?
- think about non-routine or intermittent operations.
- take account of unplanned but foreseeable events such as interruptions to the work activity.

An indicative list to assist in identifying hazards is provided in the next page:

Indicative List of Hazards or, Stressors – provided for guidance purpose only

CATEGORY	CONSIDER	CATEGORY	CONSIDER	CATEGORY	CONSIDER
Access & Egress from work place	☐ Access & egress ☐ Illumination ☐ Isolations ☐ High Voltage ☐ Ventilation ☐ Ionizing ☐ Surface Disturbance (Excavation/Demolition) ☐ Radiation	Manual Handling (Ergonomic) / Human / Muscular	☐ Repetition / Overuse ☐ High / Low reach ☐ High force / Heavy loads ☐ Awkward / Unbalanced loads ☐ Pushing / Pulling / Twisting ☐ Carrying & walking ☐ Over exertion / fatigue ☐ Design / Layout ☐ Personal characteristics e.g. height	Mechanical	☐ Unguarded moving parts (crush) ☐ Drawing in / cutting points (nips) ☐ Impact and crushing areas ☐ Uncontrolled movement ☐ Tearing / Shearing (abrasions) ☐ Unsafe access ☐ Auto-start equipment ☐ Inadequate isolation points ☐ Hand & power tool condition ☐ Stored energy (e.g. vessels) ☐ Failure of plant (loss of load, plant ejection, collapse, fragmentation)
Emergency	☐ Working alone / Remote work				
Fire / Explosion	☐ Inappropriate chemical storage/use ☐ Self-ignition combustibles (e.g. dust) ☐ Fire (e.g. burns)	Thermal	☐ Steam / Condensate ☐ Hot or cold materials/surfaces ☐ Heat stress / Cold		
Fall from height <i>Injured by people or objects falling</i>	☐ Working at height / Unguarded edge ☐ Struck by falling/lowering object ☐ Rolling/Sliding objects ☐ Ascending/Descending stairs / ramp / ladders / platform / mobile equipment ☐ Lifting equipment (e.g. crane, sling) ☐ Holes/Gaps or Slip/Trip hazards ☐ Scaffolding ☐ Unbalanced ladders ☐ Structural failure / Exceeding load ratings	Chemical / Environment <i>Toxic Corrosive Explosive</i>	☐ Hazardous Subs/Dangerous Goods ☐ Inadequate storage/bunds/labelling ☐ Gas / Dust / Fumes ☐ Poor ventilation ☐ Inhalation / Absorption / Skin contact ☐ Waste/Contamination e.g. soil/water/air	Pressure	☐ Compressed gases ☐ Hydraulic / Vacuum ☐ HP steam / Water
		Procedural	☐ Inadequate training / experience ☐ Incorrect equipment selection / use	General Work Area / Housekeeping	☐ Time of day and illumination/glare ☐ Wet / Slippery / Uneven surfaces ☐ Weather e.g. sun, wind, rain ☐ Unauthorized personnel entry ☐ Noise / Vibration ☐ Biological – fluids, insects ☐ Inadequate maintenance/inspections ☐ Restricted work area ☐ Any foreseeable abnormal conditions? ☐ Any other hazardous conditions?
		Kinetic Energy / Mobile Plant / Pedestrians	☐ Road conditions ☐ Traffic / crowds ☐ Vehicle instability / rollover ☐ Exceeding rated capacity ☐ Obstructed / Poor Visibility ☐ Being hit by the activities of another person, moving vehicle or object		
Electrical <i>Hazards resulting in electrocution / burns</i>	☐ Cables: Unsafe condition/location ☐ No earth leakage protection ☐ High voltage / Switch rooms				
Radiation	☐ Ionizing & ☐ Non-ionizing radiation	Human	☐ Untrained workers ☐ First time workers ☐ Working hours ☐ Psychological stressors ☐ Biological stressors		

Estimate of Harm / Consequence

A numerical estimate of the harm or consequence can be obtained by reference to the grid given below, example cases are mentioned for guidance.

NATURE OF HARM / SEVERITY OF CONSEQUENCE	EXAMPLE CASE	CATEGORY	Severity / Consequence RATING
First aid injury, without lost time.	a minor scratch, bruise, cut or slight burn	Insignificant	1
Loss Time injury (LTI)*	a significant cut, sprain, strain or burn	Minor	2
Hospitalization/ serious temp. disability / minor permanent disability.	burn(s), fracture(s), minor amputation, temporary disabling injury	Moderate	3
Hospitalization (longer-term)/ single death/ permanent disabilities upto 5 persons	single fatality, serious amputation, permanent disability up to 5 persons	Major	4
Multiple fatalities / permanent disability for more than 5 persons.	multiple fatalities / permanent disabilities to more than 5 persons	Catastrophic	5

*LTI refers to Lost Time Incident/Accident as defined by law.

While this grid has been constructed to consider harm to people it would be equally valid to include damage to plant / apparatus or. to the environment.

EXAMPLE CASES: RATING – CONSEQUENCE / SEVERITY / HARM

HAZARD	1 (INSIGNIFICANT)	2 (MINOR)	3 (MODERATE)	4 (MAJOR)	5 (CATASTROPHIC)
Fire	-	-	-	-	Fire/Explosion risk will always be catastrophic
Ionizing Radiation	-	-	-	-	Radiation hazard will always be considered as of extreme severity
Fall from height	-	-	-	Fall from Height less than 6 feet; leading to at least a LTI	Fall from Height more than 6 feet will always be considered as of extreme severity; leading to a Fatal accident
Worker Competency	Experienced & trained workers; no risk	Trained workers with No Experience; near miss	Experienced workers but new to this task; near miss, LTI	---	No experience, No training; fatal accident
High noise <i>refer to OD/OGN for guidance</i>	No impairment - No or very slight hearing problems. Able to hear whispers.	Slight impairment -Able to hear& repeat words spoken in normal voice at 1m.	Moderate impairment - Able to hear and repeat words spoken in raised voice at 1 meter.	Severe impairment - Able to hear some words when shouted into better ear.	Profound impairment incl. deafness - Unable to hear and understand even a shouted voice
Illumination	Higher than adequate lighting as per statutory requirements	Adequate for task; meeting legal requirements	Inadequate for task; may just be meeting legal requirements; may lead to Incident; non-LTI	Inadequate working ambience, not meeting legal requirements; may lead to LTI / permanent disability	Poor visibility leading to dangerous condition & not meeting legal requirements; may lead to fatal accident
Extreme heat and cold	No discomfort	Minor discomfort, fluid loss; more work breaks.	Discomfort, fluid loss, may lead to Incident; non-LTI.	Great discomfort, fainting; possible LTI	Impossible to work even with PPEs may lead to fatal accident

EXAMPLE CASES: RATING - CONSEQUENCE / SEVERITY / HARM (continued)

HAZARD	1 (Insignificant)	2 (MINOR)	3 (MODERATE)	4 (MAJOR)	5 (CATASTROPHIC)
Route for driving	Permanent road	Temporary road	Route with high tension wires, deep trenches etc.	Partly damaged/temporary road with high tension wires, deep trenches etc.	Damaged road / No road
Slip, trip, or fall	Unsafe Condition	Incident; non-LTI	LTI / permanent disability	Single fatality / Permanent disability for 2 or more persons	Multiple fatalities / permanent disabilities to more than 5 persons
Hazardous chemicals stressors	Exposure within permissible limit	Exposure close to permissible limit; non-LTI	Exposure crossing permissible limit; ,LTI; non-permanent disability	High exposure; Permanent disability	Very High Exposure; Multiple permanent disabilities / fatalities
Contact with mechanical energy source	First aid injury, without lost time.	Loss Time injury (LTI)	Hospitalization/ serious temp. disability / minor permanent disability.	Hospitalization (longer-term)/ single death/ permanent disabilities upto 5 persons	Multiple fatalities / permanent disability for more than 5 persons.
Contact with electrical energy source	First aid injury, without lost time.	Loss Time injury (LTI)	Hospitalization/ serious temp. disability / minor permanent disability.	Hospitalization (longer-term)/ single death/ permanent disabilities upto 5 persons	Multiple fatalities / permanent disability for more than 5 persons.
Stuck by or, against (this includes fall of objects from height)	Near Miss.	Incident; non-LTI	LTI / permanent disability	Single fatality / Permanent disability for 2 or more persons	Multiple fatalities / permanent disabilities to more than 5 persons
Ventilation	No discomfort	Minor discomfort	Discomfort, may lead to Incident; non-LTI.	Great discomfort - need additional ventilation; possible LTI / permanent disability	Dangerous to work may lead to fatal accident

EXAMPLE CASES: RATING - CONSEQUENCE / SEVERITY / HARM (continued)

HAZARD	1 (Insignificant)	2 (MINOR)	3 (MODERATE)	4 (MAJOR)	5 (CATASTROPHIC)
Contact with hot surfaces or materials	No potential for injury	Potential to cause first-degree burn.	Potential to cause second degree burn	Potential to cause third degree burn	Potential fatality
Vibration	Vibrations cannot be felt; does not affect work performance	Vibration with no physical discomfort	Vibration affecting performance without frequent breaks	Vibration with physical discomfort even with breaks	Vibration endangering safety of person working
Contact with biological pathogens	No potential for injury	Potential to cause mild reversible infection	Moderate effects, reversible disease	Severe effect long term or irreversible disease	Potential fatality
Environment Indicators	No impact	Short-term or temporary negative impact on the environment or where there is unnecessary wastage of a natural resource.	Potential to result in significant pollution, detriment to the environment and needs internal intervention.	Potential to result in public danger, which usually should be reported to the authorities	widespread, long-term, persistent negative impact on environment. Carries a high-risk of legal liability.
Invasive species	Short-term or temporary negative impact on the environment or unnecessary wastage of a natural resource.	Short-term or temporary negative impact on the environment or unnecessary wastage of a natural resource.	Potential to result in significant pollution, detriment to the environment and needs internal intervention.	Potential to result in public danger, which usually should be reported to the authorities	widespread, long-term, persistent negative impact on environment. Carries a high-risk of legal liability.

Existing Control Measures

When estimating either the Likelihood of the Harm / Consequence materializing in to danger, due account must be taken of existing control measures i.e. all the existing measures currently in place to control risks from hazards associated with the assessment must be identified. The estimates should be made assuming that the available control measures are applied. As guidance, an indicative list of such existing control measures is listed below:

- Application of E&M Safety Rules
- Local Management Instructions
- Training and Competency of staff/workers
- Work Instructions / Method Statements
- Level of Supervision
- Existing Risk Assessments (e.g. Noise, Dust, Illumination).

Estimating the Likelihood

Likelihood of some hazard materializing into harm is estimated next. Historical experience from the project, other projects in the company, other companies in the sector may be included for consideration to guide the Risk Assessor in assigning rating to such likelihood.

A numerical estimate of likelihood can be obtained by reference to the table such as the one given below.

Historical experience within/outside the project/company	Estimated Probability:	CATEGORY	RATING of LIKELIHOOD
Rare, Exceptional occurrences	Less than 1% chance of occurring.	Rare	1
Once in 3 years;	1 in 100 chance; Could occur at some time	Unlikely	2
Yearly;	1 in 20 chance; Might occur at some time	Possible	3
Monthly	Will probably occur in most circumstances	Likely	4
Weekly	Expected to occur in most circumstances	Almost Certain	5

Estimate the Risk

A numerical value for Risk can be obtained by multiplying the estimated value for likelihood by that for harm / consequence.

$$\text{Risk} = (\text{Rating of Likelihood}) \times (\text{Rating for Severity of Harm (or Consequence)})$$

It is important to understand that this numerical value of Risk has no significance in 'absolute' terms and can only be used for making relative comparisons with other assessments undertaken using this procedure.

Based on the rating the Risk may be categorized as Low, Medium or, High.

RISK SCORE EVALUATION MATRIX

RISK CALCULATOR				First aid injury, without lost time.	Loss Time injury (LTI)	Hospitalization/ serious temp. disability / minor permanent disability.	Hospitalization (longer-term)/ single death/ permanent disabilities up to 5 persons	Multiple fatalities / permanent disability for more than 5 persons.
		CONSEQUENCE / SEVERITY						
		RATING Qualitative		Insignificant	Minor	Moderate	Major	Catastrophic
Historical:	Probability:		SCORE	1	2	3	4	5
Rare, Exceptional occurrences	Less than 1% chance of occurring.	LIKELIHOOD	Rare 1	1	2	3	4	5
Once in 3 years;	1 in 100 chance; Could occur at some time		Unlikely 2	2	4	6	8	10
Yearly;	1 in 20 chance; Might occur at some time		Possible 3	3	6	9	12	15
Monthly	Will probably occur in most circumstances		Likely 4	4	8	12	16	20
Weekly	Expected to occur in most circumstances		Almost Certain 5	5	10	15	20	25



By now, the following have been completed:

1. Break the job down into successive steps or tasks
2. Identify the hazards associated with each step
3. Identify existing controls for each of the hazard
4. Estimate the potential consequences of an incident associated with each hazard
5. Carry out risk assessment.

On completion of the above, the following two steps complete the process:

- Identify possible additional controls needed for the hazards.
- Re-evaluate to check effectiveness of additional controls

Possible Additional Controls to mitigate Risk

Additional controls where required, should be prioritized with suggested hierarchy as below, mitigate the risk level. There may be a short-term (temporary) and a long-term (permanent) method of controlling the risk.

The principles to be followed are:

LONG TERM

- Eliminate hazard at source ex. modify a design.
- Automation.
- Reduce Exposure by removing 'people' from hazardous situation
- Contain hazard by enclosure

SHORT TERM

- **Engineering control** methods could include any one or, combination of the following
 1. Isolation
 2. Shielding or protective barriers
 3. Ventilation
 4. Mechanical assistance in place of manual handling
 5. Guard, enclosure
 6. Additional safety cut offs / trip systems
- **Substitution** – substitute a less hazardous material or, reduce the system energy (ex.lower the force, voltage, pressure, temperature, etc.)
- **Administrative controls** could include any one or a combination of the following
 1. Permit to work system
 2. Personal Supervision

- | | |
|---------------------------|------------------------------|
| 3. Work scheduling/timing | 5. Job rotation |
| 4. Access controls | 6. Signage, warnings, lights |

- **Personal Protective Equipment (PPE)** could include any one or a combination of the following:

- | | |
|-------------------------------------|---------------------------------|
| 1. Gloves | 6. Safety glasses / face shield |
| 2. Respirator | 7. High-visibility jackets |
| 3. Arc flash suits | 8. Long sleeves |
| 4. Thermal shielding | 9. Coveralls |
| 5. Safety belts/nets/fall arrestors | |

- **Emergency Response /Plant /Equipment / Access**

- | | |
|--|--------------------------------------|
| 1. First aid equipment/Trained firstaiders | 5. Vehicular type / access / parking |
| 2. Fire equipment | 6. Signage / Barricades |
| 3. Other emergency response | 7. Communication equipment |
| 4. Chemical/Oil spill kit | 8. Emergency Rescue arrangements |

- **Training / Competency**

- | | |
|--------------------------------|---------------------------------|
| 1. Certificates of competency | 5. Any relevant legislation |
| 2. Experience | 6. Consultation / Notifications |
| 3. Induction training required | 7. Consider literacy aspect |
| 4. Project / Area Supervisor | |

In applying the hierarchy consideration should be given to the relative costs, risk reduction benefits, and reliability of the available options.

Recording and documenting the risk assessment results

Records shall be maintained as required under [Clause 7.0 Records](#).

Annexure 2 – Hazard Identification & Risk Assessment Form

STEP 1.							
Work / Task:		Work Type:	Routine / Non-Routine / Emergency	Project/ Dept.		Number	
Work / Task Description:				EIC			

STEP 2.		STEP 3.	STEP 4.	STEP 5.	STEP 6.	STEP 7.	STEP 8.	STEP 9.			STEP 11.
No.	SUB-TASKS / ACTIVITY STEPS	IDENTIFY POTENTIAL HAZARDS	EXISTING CONTROL MEASURES	RISK RANKING			ADDITIONAL RISK CONTROL MEASURES - Hierarchy of Control – Elimination, Substitution, Isolation, Engineering, Administration, Personal Protection. - Additional information can be attached.	RESIDUAL RISK			Action To Be Taken by/ Initials
				Consequence	Likelihood	Risk Rating		Consequence	Likelihood	Residual Risk	
STEP 9.							Highest Remaining Residual Risk				

STEP 10.			
Risk assessment prepared by: (all members, if applicable)		Date:	/ / 20__
Consultation conducted with:(if applicable)		Work can Proceed?	Yes / No

Residual Risk	Step 12. APPROVAL REQUIRED TO PROCEED	Approved by:	Approval Date:
Extreme (E)	Approval must be obtained from the HOP before work starts		
High (H)	Approval must be obtained from the Head (O&M)/ Head (Project) before work starts		
Medium (M)	Approval from HOS or Section Head has approved this risk assessment, & all identified control measures are in place.		
Low (L)	Work can be commenced by EIC when all identified control measures are in place. EIC can sign-off this risk category.		